

Data Binding in Angular (20)

1. What is Data Binding in Angular?

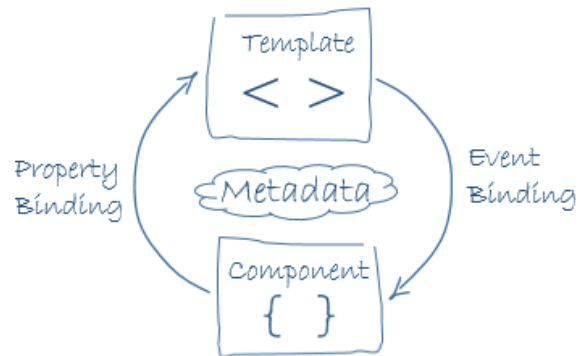
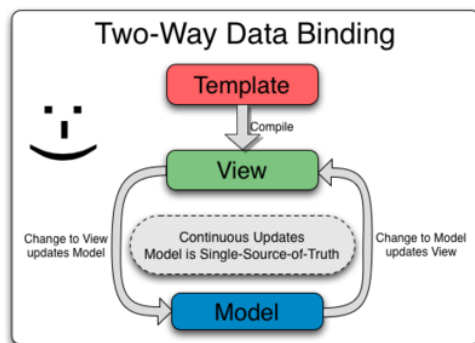
Data Binding is the mechanism that synchronizes data between:

- **Component class (TypeScript / State)**
- **Template (HTML / View)**

Angular's data binding ensures that:

- Changes in the **component state** are reflected in the **UI**
- User actions in the **UI** update the **component state**

Angular uses a **unidirectional data flow by default**, with explicit mechanisms for two-way binding.



2. Why Data Binding is Important?

Without data binding:

- You would manually query the DOM
- Update elements using JavaScript
- Handle synchronization yourself

With Angular data binding:

- UI updates automatically

- Code is declarative and predictable
 - Separation of concerns is maintained
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3. Types of Data Binding in Angular (20+)

Angular supports **four types of data binding**:

Type	Direction
Interpolation	Component → View
Property Binding	Component → View
Event Binding	View → Component
Two-Way Binding	Component ↔ View

Let us examine each in detail.

4. Interpolation ({{ }})

Purpose

Display component data inside the template.

Syntax

```
{{ expression }}
```

Example (Angular 20+ Standalone)

Component

```
@Component({  
  selector: 'app-user',  
  standalone: true,  
  template: `<h2>Welcome, {{ userName }}</h2>`  
})  
  
export class UserComponent {
```

```
    userName = 'San';  
}
```

Key Points

- Only **read-only**
 - Cannot perform assignments
 - Supports expressions, not statements
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5. Property Binding ([property])

Purpose

Bind a **DOM property** to a component value.

Syntax

```
[property]="expression"
```

Example

```
<img [src]="imageUrl" [alt]="imageAlt">
```

```
<button [disabled]="isDisabled">Save</button>
```

```
imageUrl = 'logo.png';
```

```
imageAlt = 'Company Logo';
```

```
isDisabled = true;
```

Difference from HTML Attributes

Attribute	Property
Static	Dynamic
String only	Any data type
Evaluated once	Continuously updated

6. Event Binding ((event))

Purpose

Capture user actions and notify the component.

Syntax

```
(event)="handler()"
```

Example

```
<button (click)="increment()">+</button>
```

```
<p>Count: {{ count }}</p>
```

```
count = 0;
```

```
increment() {  
  this.count++;  
}
```

Common Events

- click
- input
- change
- submit
- keyup

7. Two-Way Data Binding ([[]])

Purpose

Synchronize data **both ways**.

Syntax

```
[(ngModel)]="property"
```

Example (FormsModule Required)

```
@Component({
```

```
standalone: true,  
imports: [FormsModule],  
template: `  
  <input [(ngModel)]="name">  
  <p>Hello {{ name }}</p>  
`,  
})  
export class TwoWayComponent {  
  name = 'Angular';  
}
```

How It Works Internally

`[(ngModel)]="name"`

is syntactic sugar for:

`[value]="name"`

`(input)="name = $event.target.value"`

Recommendation (Angular 20+)

- Use **Reactive Forms** or **Signals** for large applications
 - Use `ngModel` mainly for simple forms
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9. Attribute vs Class vs Style Binding

Attribute Binding

`<td [attr.colspan]="colSpan"></td>`

Class Binding

`<div [class.active]="isActive"></div>`

Style Binding

```
<div [style.color]="isError ? 'red' : 'green'"></div>
```

10. Binding with Structural Directives (Angular 20 Syntax)

Conditional Rendering

```
@if (isLoggedIn) {  
  <p>Welcome back</p>  
} @else {  
  <p>Please login</p>  
}
```

Loop Binding

```
@for (item of items; track item.id) {  
  <li>{{ item.name }}</li>  
}
```

11. One-Way vs Two-Way Data Flow (Best Practice)

Approach	Recommendation
One-way binding	Preferred
Event-driven updates	Recommended
Two-way binding	Use sparingly
Signals + Events	Best practice in Angular 20

12. Common Data Binding Mistakes

- Using interpolation inside property binding
- Modifying state directly in templates

- Overusing ngModel
 - Forgetting to import FormsModule
 - Calling heavy functions inside {{ }}
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13. Summary

- Angular supports **four data binding types**
 - Templates remain **declarative**
 - Business logic stays in **components**
 - Data binding is the foundation of Angular UI architecture
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